Manish Pandey

Backend Developer

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LinkedIn | GitHub | LeetCode | Portfolio

SUMMARY

Backend developer with hands-on experience building production-grade distributed systems, microservices, and real-time applications. Proficient in Node.js, NestJS, TypeScript, MongoDB, and event-driven architectures, with growing exposure to Java, CI/CD, and cloud tooling through current work at TCS, alongside a continued interest in Golang for high-performance tooling. Passionate about writing clean, scalable code and continuously exploring new technologies to solve meaningful engineering challenges.

SKILLS

Languages: TypeScript, JavaScript (ES6+), Java, C/C++, Golang (Basic)

Backend: Node.js, NestJS, Express.js, Mongoose, REST APIs, GraphQL, BullMQ, json-rules-engine

Databases: MongoDB (Aggregation, Geospatial, Indexing), PostgreSQL, Redis, Firebase

Real-time: WebSockets, Redis Pub/Sub, RabbitMQ

Cloud & Infra: AWS S3 (Pre-signed URLs), CI/CD, GitLab, Linux, Cron Scheduling, PM2

Tools: Git, GitHub, GitLab, Postman, Apidog, Jest, VS Code

PROFESSIONAL EXPERIENCE

Assistant System Engineer Trainee

21 May 2026 – Present

Tata Consultancy Services (TCS) | *India*

- Onboarded into TCS's cloud engineering track, building hands-on proficiency in Java, CI/CD pipelines, and version control workflows using GitLab and GitHub.
- Learning core cloud engineering practices including automated build/deploy pipelines and collaborative source-control workflows, laying the foundation for enterprise-scale, cloud-native development.

Backend Developer

11/2024 – 17 April 2026

Katyayani Organics | *Bhopal, India*

- Engineered an automated shipment tracking pipeline processing 36,000+ orders per cron cycle across 4 courier partners (Delhivery B2C/B2B, Shiprocket, Velocity) using concurrent batch workers, rate-limit handling with Retry-After parsing, and canonical status mapping across 10 delivery states — reducing manual tracking overhead by 90%.
- Optimized order fulfillment by designing a MongoDB 2D-sphere geospatial routing algorithm for warehouse selection based on inventory availability and proximity — reducing shipping costs by 21% and eliminating external map API dependencies.
- Built an event-driven webhook engine supporting configurable triggers with AND/OR conditional evaluation, dynamic dot-notation payload mapping, JSON schema validation, dual-mode authentication (static headers / OAuth2), and automatic retry — powering cross-service communication across 6 microservices.
- Developed a payment reconciliation system (UTR tracking) enforcing accounting invariants, bidirectional auto-verification between orders and transactions, multi-bank UTR format validation (Razorpay, ICICI, HDFC, ZOHO), and 10 automated data-integrity checks with Slack alerting.
- Designed a rule-based promotional engine using json-rules-engine with nested AND/OR conditions, real-time inventory checks, agent distribution quotas with period-based auto-reset, customer recurrence limits, and priority-based rule evaluation across 40+ configurable fact parameters.
- Built real-time monitoring dashboards using WebSockets and Redis Pub/Sub with Node.js clustering and sticky sessions — powering live inventory tracking, agent performance leaderboards, and critical event notifications across distributed server environments.

Assistant Art Director

07/2024 – 10/2024

Lake Town | *Bhopal, India*

- Created and executed visual concepts for client projects; edited videos using Adobe Premiere with audio synchronization and storytelling.
- Captured and optimized photo/video content for platforms like Instagram, ensuring high visual quality and platform compatibility.

PROJECTS

Order Routing & Fulfillment Optimization

Tech Stack: *NestJS, TypeScript, MongoDB, REST APIs*

- Built a smart order routing system using NestJS and TypeScript that automatically picks the nearest warehouse with available stock to fulfill each order, reducing shipping costs and delivery time.
- Designed an order-splitting mechanism that breaks large orders across multiple warehouses when a single location doesn't have full inventory.
- Added an inventory hold feature at the time of order placement to prevent overselling when multiple orders come in at the same time.

Automated Shipment Tracking Pipeline

Tech Stack: *Node.js, MongoDB, REST APIs, S3, Slack*

- Built an automated tracking system that fetches live shipment updates from multiple courier partners at regular intervals and keeps all order statuses up to date.
- Handled duplicate updates and overlapping runs gracefully so the system stays consistent even under heavy load.
- Added smart retry logic for when courier APIs hit rate limits or go temporarily unavailable.
- Set up automatic return-to-origin detection that notifies the team on Slack and triggers order cancellation in the CRM.

Payment Reconciliation System (UTR Module)

Tech Stack: *Node.js, Express, MongoDB, Razorpay API, Slack*

- Built a payment tracking system that automatically matches incoming payments to their corresponding orders, eliminating the need for manual reconciliation.
- Supports multiple banks and payment formats with automatic validation and verification through Razorpay.
- Runs automated integrity checks to catch issues like duplicate payments, mismatched amounts, or unlinked transactions, with alerts sent to Slack.
- Includes bulk CSV upload for processing large batches of payments and generating reports.

Event-Driven Webhook & Integration Engine

Tech Stack: *Node.js, MongoDB, PostgreSQL, REST APIs*

- Built a flexible webhook system that lets different services communicate with each other through configurable event triggers and conditions.
- Supports dynamic payload mapping, request validation, and multiple authentication methods so new integrations can be added without writing code.
- Includes detailed logging and manual retry options for easy debugging when something goes wrong.

EDUCATION

B.Tech in Computer Science | 2021 – 2025 | Bhopal, India
Sagar Institute Of Research And Technology

Secondary and Higher Secondary | Bhopal, India
Mother Teresa Sr. Sec. Co-Ed School

INTERESTS

Playing Chess • Mixed Martial Arts • Learning About New Technologies • Binge-watching Anime